SASEUL ORIGIN

ArtiFriends

Contents

1. Introduction
   1.1 What is SASEUL
   1.2 Disclaimer
2. Terminology
   2.1 Node
   2.2 Network
   2.3 Block data
   2.4 Generation
3. Consensus - HAP-2 Algorithm
   3.1 Network Lifecycle
   3.2 Transaction Verifying - Smart Contract
4. Definition of SASEUL Nodes
   4.1 Light
   4.2 Validator
   4.3 Supervisor
   4.4 Arbiter
5. Requirements

1. Introduction

1.1 What is SASEUL

Blockchain is, an immutable time-stamped series record of data that is distributed and managed by a cluster of computers.

In a simple phrase, blockchain (originally block chain) is a growing list of records, called blocks, that are linked using cryptography. Each block contains a cryptographic hash of the previous block, a timestamp, and transaction data (generally represented as a Merkle tree).

By design, blockchain is resistant to modification of data. It is "an open, distributed ledger that can record transactions between two parties efficiently and in a verifiable and permanent way". To implement a *distributed ledger*, blockchain is typically managed by a peer-to-peer network, where participants collectively adhere to a protocol for inter-node communication and block validation. Once recorded, data in any given block cannot be altered retroactively without alteration of all subsequent blocks, which requires consensus of the network majority. Blockchain records are not *impossible to* alter. However, blockchain technology is inherently designed to boost security and exemplify a distributed computing system with high Byzantine fault tolerance. Decentralized consensus is therefore a perfect fit to implement inter-node communication within the blockchain network.

For these reasons, many industry leaders expect to achieve significant business benefits, including greater transparency, enhanced security, improved traceability, increased efficiency and speed, and reduced costs.

Unfortunately, there are no blockchain technologies out there that can satisfy the industrial standards. To surpass industrial standards, blockchain technology needs three features - decentralization, stability and fast speed (transactions per second). All existing blockchains lack at least one of them. However, SASEUL is the first blockchain technology that holds all three values - decentralization with fast tps and stability. SASEUL is proving that it satisfies industrial standards through a lot of use cases.

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2. Terminology

2.1. Node

Node is a machine just like a computer or cell phone, which has Memory, Storage(HDD, SDD) and CPU (or GPU) including communication protocol and equipment between nodes.
Node has only one Network Address and one Key Address. A person can own multiple nodes. Distributed processing system like Load Balancer allows many nodes to operate like a single node.

2.2. Network

Network in this whitepaper is basically the Blockchain Network. Blockchain Network is the group of nodes sharing same block data and should generate same transactions and decisions. If nodes make different blocks or decisions, the node will be in a different network. This process is called a 'Fork'.

2.3. Block data

Block data is all the data shared by Blockchain Network and consists of Status data, Transaction data, Decision data and Hash data.

Status data is the dataset of databases in general internet service. Each node changes the status according to the result of transaction handling.

Transaction data is a request made to the network to change the status data. Nodes process transactions according to a set rule. Accepted transaction changes the status, denied transactions do not change the status.

Blockhash is created using decision data and transaction data, and each data is connected using the hash of previous block together.

2.4. Generation

Certain nodes compress the block data when a sufficient number of block data accumulates. Each compressed block is called a Generation Block.

3. Consensus - HAP-2 (or POR)

SASEUL has invented Proof of Rule (POR) consensus algorithm without BFT issue and HAP(Hypothesis Acceptance Protocol), the piece of software, was implemented with this consensus. HAP-2 is the second version of HAP and HAP-2 is used synonymously with POR.

The core idea of HAP-2 is that all nodes of SASEUL are very selfish decision makers, who focus on building their own blocks. A node builds a hypothetical block after gathering and merging all the transactions from other nodes and rearranging them by timestamp. The node compares this hypothetical block with others' blocks and finally decide to attach it selfishly.

3.1. Network Lifecycle

Background Activity: Sync
A node that has shorter blocks than others, just pull the extra blocks from a node that has the longest blocks and sync the extra blocks one by one after verifying them. After this process, the node begins the process below.

a) Ping
   Nodes periodically collect information about status from others

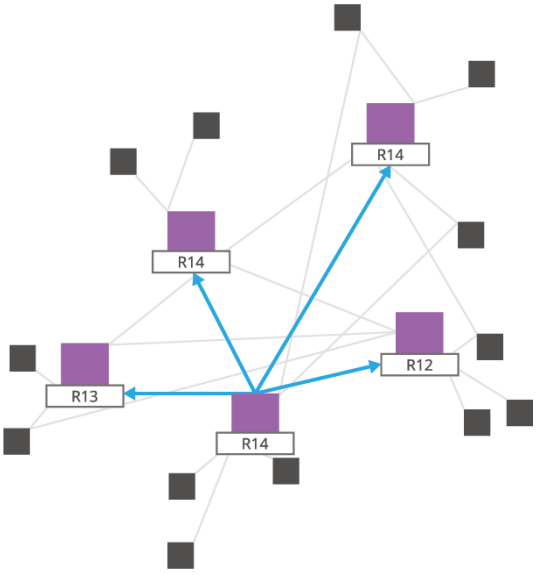

Figure 1. Collecting status data.

b) Receive a transaction
Nodes(Validators) participating in consensus collect transactions requested from the network.

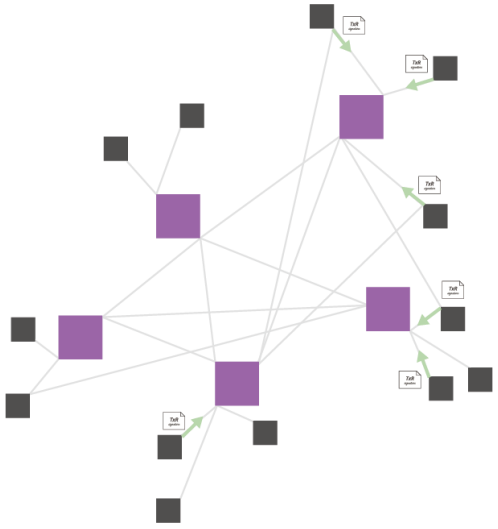

Figure 2. Receive a transaction

c) Rounding
The node that collects transaction creates a Hypothesis (data Chunk) to become the next block.

d) Data pulling
After completing the c) Rounding phase, Nodes collect the network hypotheses.
Collecting hypotheses is a process that starts when nodes find hypotheses of other nodes during the a) Ping phase.
Nodes must ensure that their hypotheses do not reverse the decision by putting their Signature on the hypothesis they present.

e) Make a decision
Collected hypotheses in d) step are disassembled and the disassembled transactions are rearranged by timestamp and other factors. Approval or disapproval of the respective transaction is decided by verification. After all these processes, Blockhash will be generated.

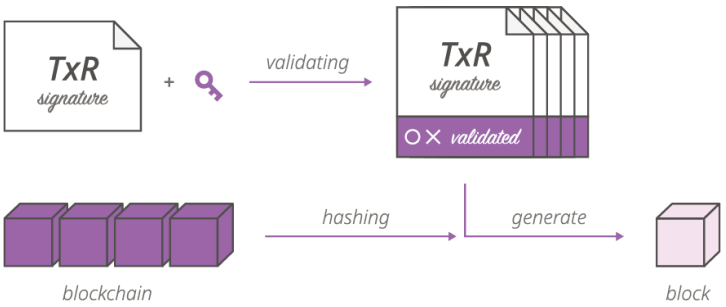

Figure 3. Building Blockhash with previous Block New Transaction and Decision Data.

f) Pre-commit
Block is created with the accepted hypotheses in d) step and Blockhash in e) step. The node will then compare the Block with others' Blocks . Depending on the result of comparison, the node decides whether it merges the Block or Fork the Node based on several rules.

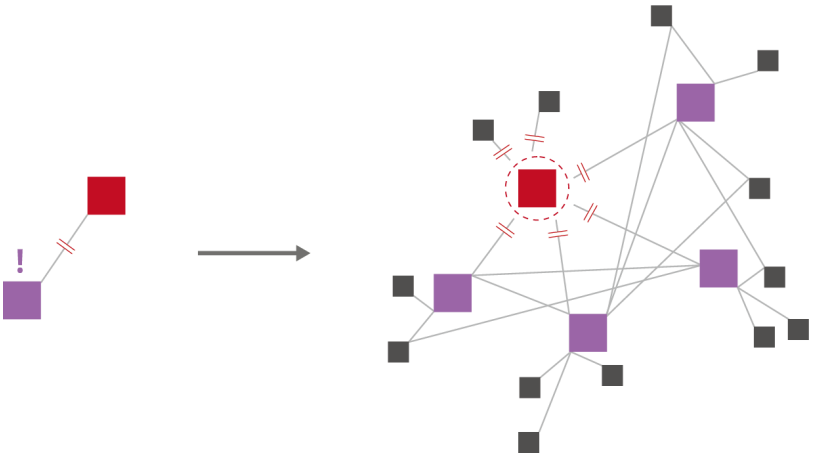

Figure 3. The node which has the different decision, is considered as Evil node.

g) Commit
Confirm Block and repeat from a) step.

3.2. Transaction Verifying - Smart Contract

All the Transactions are definitely controlled by a Smart Contract. Contract Code follows the 4 steps below.

1. Check transaction structure & validity
   Check the structure and speculation of requested Transaction.
   Transaction should include all the variables set accordingly and have the signature created by Private key of Requestor.

2. Load status
   Get the Status from Block data for handling Transaction.

3. Make a decision
   Decide the 'Accept/Deny' depending on Status.

4. Set status
   If accepting the Transaction, change the Status and SASEUL Engine basically provide below four System Contracts.

   1. Genesis
      This is the Contract to create the first Block. Attribute, Method, and Contract can be predefined. First declarator is always validator.

   2. UpdateAttribute
      This is the Contract to change the structure of Status defined by Network. The time of changing the structure could be assigned.

   3. UpdateMethod
      This is the Contract to change and define the Methods in Contract. The time of changing the structure could be assigned.

   4. UpdateContract
      This is the Contract to define new Contract or change existing Contract. The time of changing the structure could be assigned.

4. Definitions of SASEUL Nodes

4.1 Light node

Light nodes are the most basic and simple nodes on the network. They are directly related to blockchain wallet and practical services for the end users who do not actively contribute to the network. Each node has its unique private key and tracker and is capable of requesting transactions and viewing the transaction results.

4.2 Supervisor

Supervisors are nodes that sync all existing blocks on the network. They differ from light nodes because they are meant to monitor and supervise the blockchain network. The supervisors check whether the network has

Light nodes are the most basic and simple nodes on the network. They are directly related to blockchain wallet and practical services for the end users who do not actively contribute to the network. Each node has its unique private key and tracker and is capable of requesting transactions and viewing the transaction results.

4.3 Validator

Validators also sync all existing blocks on the network and are authorized to participate in the consensus process. They are capable of receiving transaction requests and either accepting or denying them based on consensus to generate data in a block form. The fundamental consensus is derived from the SASEUL Consensus Algorithm. Validators communicate directly with light nodes, periodically hash generated blocks, and transfer those blocks to the arbiters.

4.4 Arbiter

Arbiters are storage nodes for all blocks created in the network. They do not participate in the consensus process. The only responsibility they have is sending past data when a validator requests them for reference during validation.

5. Requirements

Validator, Supervisor Node
Minimum: 2 Core, 4 Thread CPU
Recommendation: 4 Core, 8 Thread CPU

Minimum: 250 GB SSD
Recommendation: 1 TB SSD

Minimum: 8GB Memory
Recommendation: 16GB Memory

Minimum: 5 Gbps Network Capacity
Recommendation: 20 Gbps Network Capacity

Arbiter
Minimum: 4 Core, 8 Thread CPU
Recommendation: 8 Core, 16 Thread CPU or GPU equivalent spec of the CPU.

Minimum: 1 TB SSD
Recommendation: The more, the better.

Minimum: 16GB Memory
Recommendation: 64GB Memory

Minimum: 20 Gbps Network Capacity
Recommendation: The more, the better.